Skills Summary

One Input, One Output

Use this reference while completing your worksheet or when studying.

Definition: a relation is a *function* when every input is paired with exactly one output. The rule is about *inputs*. Repeated outputs are always fine; a repeated *input* with two different outputs is what breaks it.

Skill 1 — Deciding from a table

Rule: scan the input row for a value that appears twice. If one does, check whether its two outputs differ — if they do, the relation is not a function.

Example: Is the table with inputs $2, 5, 8$ and outputs $7, 9, 13$ a function, and what is the output at input 5?
Step 1. The definition is about inputs, so read the input row first and only then look up the value asked for.
Step 2. The inputs 2, 5 and 8 are all different, so no input has a choice of outputs; the column under 5 gives its output.
A function, and the output at 5 is **9**.

Try it: Inputs $1, 3, 6$ with outputs $4, 4, 9$. What is the output at input 3, and is it a function?

check: **4**, and yes — repeated outputs are allowed

Skill 2 — Deciding from a list of pairs

Rule: look at the first number of every pair. If the same first number turns up twice with different second numbers, the relation is not a function.

Example: Is $(3, 2)$, $(7, 5)$, $(3, 9)$ a function? Compare the two outputs of the repeated input.
Step 1. Only the first numbers matter for the test, so list them and look for a repeat before doing anything else.
Step 2. The input 3 appears twice, with outputs 2 and 9 in that order, and those two numbers are different.
Not a function, because $\mathbf{2 < 9}$ rather than equal.

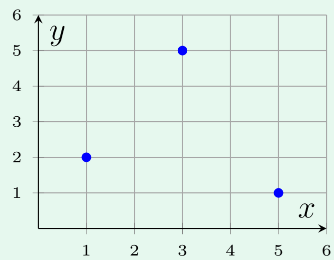
Try it: For $(8, 10)$, $(5, 6)$, $(8, 4)$, compare the two outputs of the repeated input, in the order listed.

check: $\mathbf{10 > 4}$, so not a function

Skill 3 — Deciding from a graph

Vertical-line test: if any vertical line meets the graph more than once, two points share an input and the relation is not a function. Points side by side at the same height are fine.

Example: Is the plotted relation a function, and what is the output at input 3?



Step 1. Sweep an imaginary vertical line across the picture: the question is whether it ever catches two dots at once.
Step 2. The three dots sit at three different horizontal positions, so it never does; the dot above 3 is at height 5.
A function, and the output at 3 is **5**.

Try it: A graph shows the points $(2, 3)$, $(4, 6)$, $(6, 2)$. What is the output at input 4, and is it a function?

check: **6**, and yes

Skill 4 — Relations given by a rule

Rule: a relation written as a rule is always a function. The arithmetic gives one result for each input, so an input can never come back with two different outputs.

Example: For the rule $y = 5x - 2$, find the output when the input is 4.
Step 1. A rule needs no repeated-input search — one substitution settles both the value and the fact that it is a function.
Step 2. $5(4) - 2 = 20 - 2$, and there is no other answer the arithmetic could give.
The output is **18**.

Try it: For the rule $y = 6x + 1$, find the output when the input is 5.

check: **31**